SUPERIOR UNIVERSITY

Assignment 2

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Section: 6A

Teacher: Mr. Qasim

Subject: Software construction

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Scenario:

Develop a simple Online Shopping and Payment Processing System in Python. The system should allow a customer to place an order containing different products and complete checkout using a selected payment method.

The store sells physical products (for example, a keyboard or a book) and digital products (for example, an e-book or an online course). These product types may have different pricing or order-cost considerations. Customers may pay using different methods, such as card or cash.

Python source code (.py files). (In an other file)

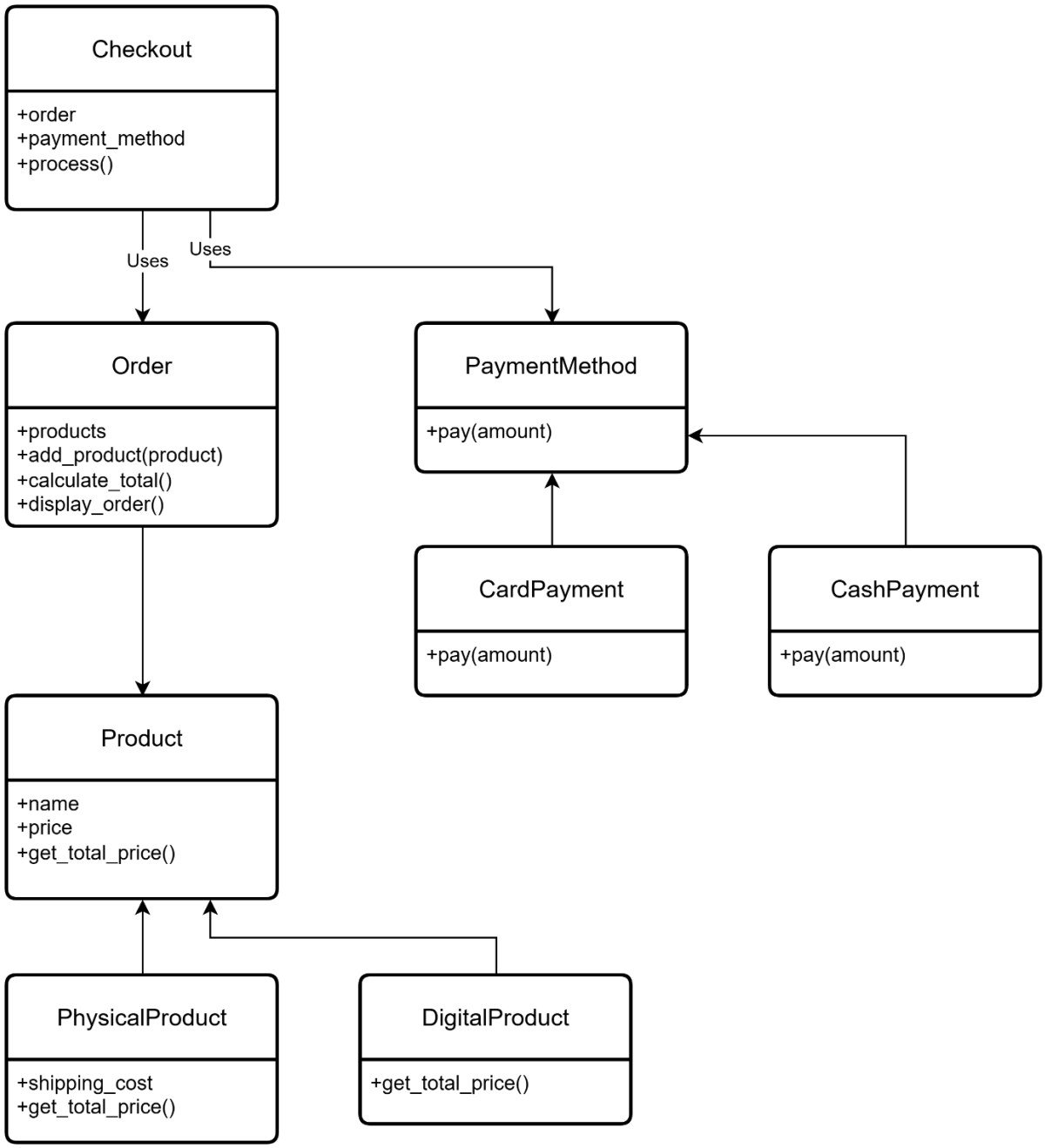
A short class diagram or clearly labelled sketch showing the main classes and their relationships. (In this file)

Screenshots or sample output demonstrating an order with more than one product and checkout using at least two payment methods. (in this file)

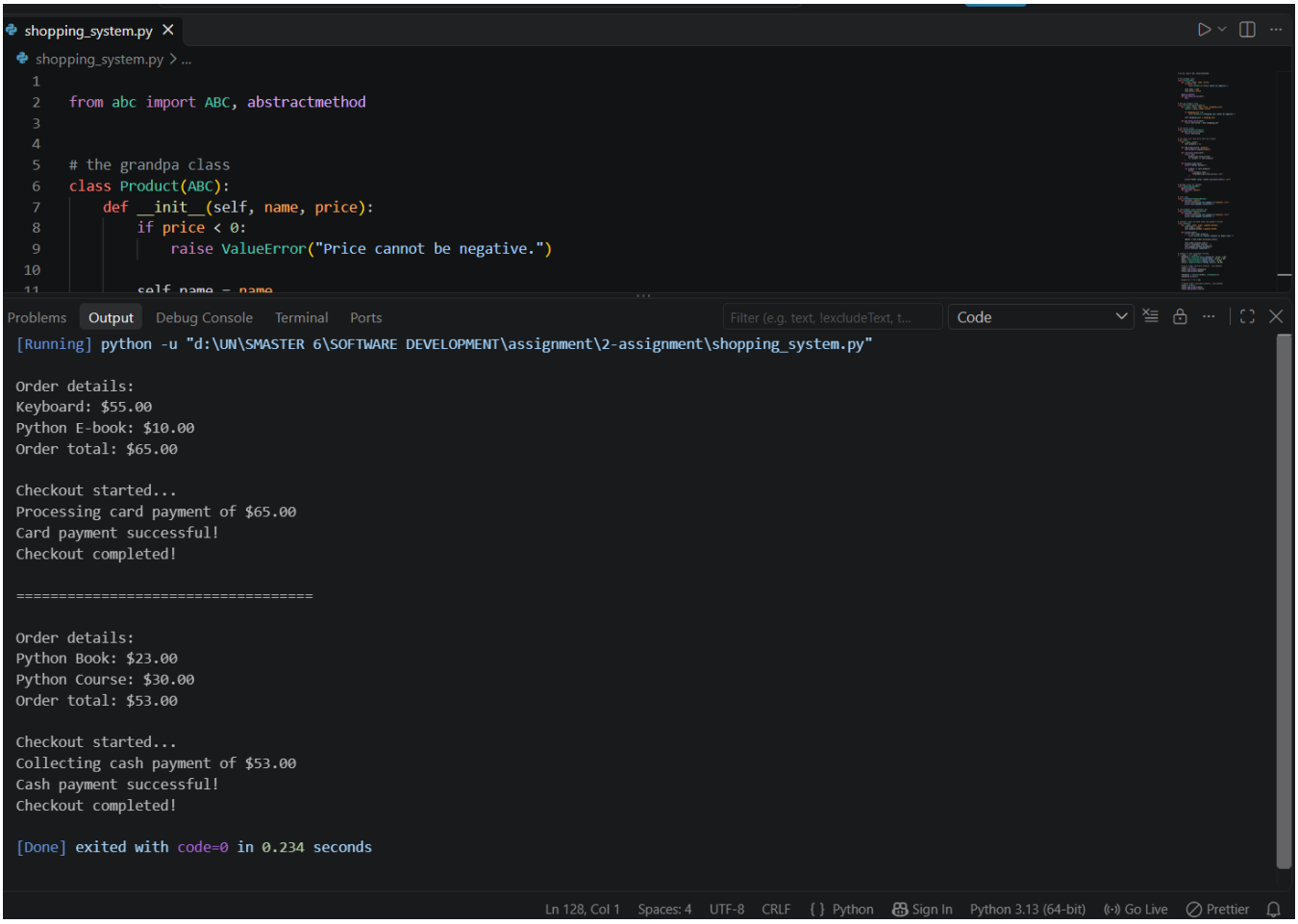
A brief explanation (approximately ½–1 page) identifying where you used inheritance, polymorphism, and composition, and why (in this file)

Answer:

class diagram:



example output:



brief explanation:

1. Inheritance

PhysicalProduct and DigitalProduct inherit from Product, and also CardPayment and CashPayment inherit from PaymentMethod.

Inheritance allows related classes to share common functionalities. For example every product has a name and price, so those attributes are defined once in Product instead of being duplicated in both product classes. The payment classes share the requirement to implement pay().

2. Polymorphism

Polymorphism appears in two places: get_total_price() and pay().

Both product types implement get_total_price(), but a physical product includes shipping while a digital product does not. Likewise, CardPayment and CashPayment implement pay() differently.

The Checkout class can call self.payment_method.pay(amount) without needing to know which payment method it received. Each subclass provides its own behavior through the same interface.

3. Composition

An Order contains a list of Product objects. A Checkout object contains references to an Order and a PaymentMethod.

These are composition-style has-a relationships: an order has products, and checkout has an order and a payment method. This separates responsibilities and lets us change the products or payment method without rewriting the checkout logic.

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Question part **Reflection**: ..... **Error! Bookmark not defined.**

Answer part **Reflection**: ..... **Error! Bookmark not defined.**

“Thank you for reviewing this assignment.”